Basic Idea - Weighted Summation

Let $g_1 \cdots g_n$ be different vectors

Let $W_1 \cdots W_n$ be weighting factors

Such that $g_f = \sum_{i=0}^{n} W_i g_i$

How do we choose our W's?

A) Practical consideration - saturate vector length (Magnitude)

1) Convex combination

$W_i \in [0\ 1]$, $\sum_{i=1}^{n} W_i = 1$

if each $\|g_i\|$ then the summed vector g will be:

$g \leq g_{max}$

2) Thresholding

Define a maximum velocity V_{max}

$V_d = \min(\|g\|, V_{max})$

$\Rightarrow g_d = \frac{V_d}{\|g\|} g$

I Attempt 1: Go-to-goal and Obstacle Avoidance

$g = \frac{V_d}{d} \text{scale}(d)(q_d - q)$ Attract

$g = \frac{V_d}{d} \text{scale}(d)(q - q_0)$ Repulse

II Attempt 2: Smarter Obstacle Avoidance

$$g = K_I \cdot K_o \cdot A \cdot \hat{q} \qquad A = \begin{bmatrix} \gamma & -\omega \\ \omega & \gamma \end{bmatrix}$$

Influence based on distance

Influence from behind

$$K_I(d) = \begin{cases} 0 & d > S \\ \frac{S-d}{S-r} & r \le d \le S \\ 1 & d \le r \end{cases}$$

$$K_o = \begin{cases} 1 & |\theta_o| \le \underline{\pi/2} \\ 0 & \text{otherwise} \end{cases}$$

Use atan2 trick

$$\omega = \begin{cases} |\omega_d| & |\theta_d| > 0 \\ -|\omega_d| & |\theta_d| \le 0 \end{cases}$$

We don't want to be attracted to the orbit, we just want to orbit away if we need to.

$$\gamma = \begin{cases} k(r^2 - \hat{q}^T \hat{q}) & d \le r \\ 0 & \text{otherwise} \end{cases}$$

III Attempt 3: More avoidance

For every point we measure we have a repulsive force & orbit. However it is probably best to have a small sphere of influence for orbit.

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